QuickSyn session report

20261014_163313_FSW-0020_0000007f

QuickSyn Session Reduction and Veusz Export 0.0.10

Session 2026-10-14 16:33:13 UTC (UTC+0000) to 2026-10-14 16:33:54 UTC (UTC+0000)

SIMULATED SESSION: no instrument was connected; every value below comes from the controller's built-in simulator.

Contents

1 Instrument **2**

2 Acquisition timing **2**

3 Generated signal **3**

4 Thermal **3**

5 Stability of the reported carrier **3**

6 Plots **5**

6.1 Overview (all quantities) . . . 5

6.2 Carrier frequency . . . 6

6.3 Output level . . . 6

6.4 Internal temperature . . . 7

6.5 Serial I/O latency . . . 7

6.6 Sample interval . . . 8

6.7 Allan deviation of the reported carrier . . . 8

7 Files in the session folder **9**

8 Provenance **9**

References **9**

1 Instrument

Field	Value
`*IDN?` reply	`Phase Matrix,FSW-0020,0000007f,1530200001,0008`
Model	FSW-0020
Option word	`0000007f` (options 1, 2, 3, 4, 5, 6, 7)
Serial number	`1530200001`
Firmware version	`0008`
Datasheet coverage	0.5 to 20 GHz
Transport	SIM at 115200 baud (simulated)

`*IDN?` fields: manufacturer, model, option word, serial, firmware. [1] does not define the order; it was confirmed against NI's soft front panels on an FSL-0020 (option bit n = option $n+1$).

2 Acquisition timing

Quantity	Value	Unit
Samples	414	
Loaded from	chunks	
Duration	41.2965	s
Requested sample rate	10	Hz
Achieved sample rate	10.0009	Hz
Mean sample interval	99.9914	ms
Interval standard deviation (jitter)	2.73932	ms
Interval minimum	78.2154	ms
Interval maximum	124.93	ms
Jitter relative to the period	27395.5	ppm
Device I/O latency, mean	5.96576	ms
Device I/O latency, 95th percentile	8.72443	ms
Device I/O latency, maximum	11.9666	ms

Timing of the telemetry stream.

The sample period is the reciprocal of the requested rate f_s,

$$\tau_0 = \frac{1}{f_s}, \tag{1}$$

and the achievable rate is bounded by the serial round trips each sample costs. With N_q queries per sample and a mean round-trip time t_q,

$$f_{s,\max} = \frac{1}{N_q\, t_q}. \tag{2}$$

The jitter is the standard deviation of the measured intervals,

$$\sigma_{\tau_0} = \operatorname{std}\left(t_i - t_{i-1}\right), \tag{3}$$

taken from the monotonic clock of the logging computer.

3 Generated signal

Quantity	Value	Unit
Distinct frequencies observed	5	
Frequency minimum	8.250000000	GHz
Frequency maximum	19.500000000	GHz
Power minimum	-3.0	dBm
Power maximum	+13.0	dBm
RF output on, fraction of samples	0.000	
Samples with RF unlocked	0	
Samples with reference unlocked	0	
Samples with voltage error	0	

Signal state reported by the instrument.

Status bits are decoded from `STAT?` with the bit map of [1, Table 5]; the lock bits are set when *unlocked*.

4 Thermal

Quantity	Value	Unit
Internal temperature minimum	29.0	°C
Internal temperature maximum	29.5	°C
Rise over the session	+0.5	°C
Margin to the +75 °C limit	45.5	°C

Internal temperature (DIAG:MEAS? 21).

The +75 °C internal limit is the caution in [2].

5 Stability of the reported carrier

The fractional frequency error of sample i against the median commanded carrier f_c is

$$y_i = \frac{f_i - f_c}{f_c}, \tag{4}$$

and the overlapping Allan deviation at averaging time $\tau = m\,\tau_0$ follows IEEE Std 1139-2008 [4]:

$$\sigma_y^2(\tau) = \frac{1}{2\,(N - 2m + 1)\,\tau^2} \sum_{k=1}^{N-2m+1} \left(x_{k+2m} - 2x_{k+m} + x_k\right)^2, \tag{5}$$

with the phase residuals

$$x_k = \tau_0 \sum_{i<k} y_i. \tag{6}$$

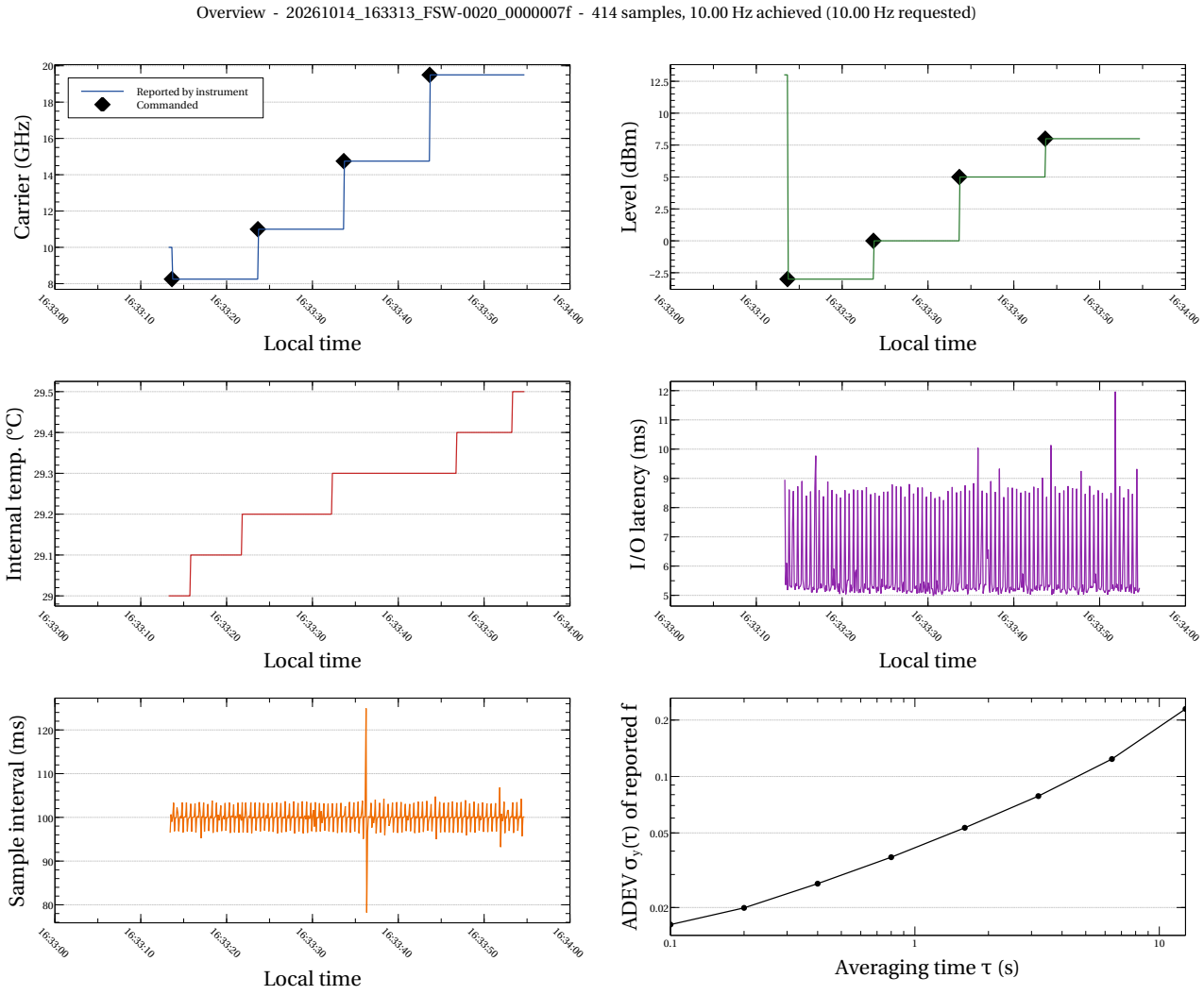
This characterises command fidelity and the logging time base, not the phase noise of the carrier: `FREQ?` returns the commanded setting, not a measurement of the output. Phase noise is specified separately in the datasheet [3] and cannot be inferred from this log.

Averaging time τ (s)	**ADEV $\sigma_y(\tau)$**
0.09999	1.622e-02
0.2	1.991e-02
0.4	2.681e-02
0.7999	3.710e-02
1.6	5.325e-02
3.2	7.859e-02
6.399	1.239e-01
12.8	2.292e-01

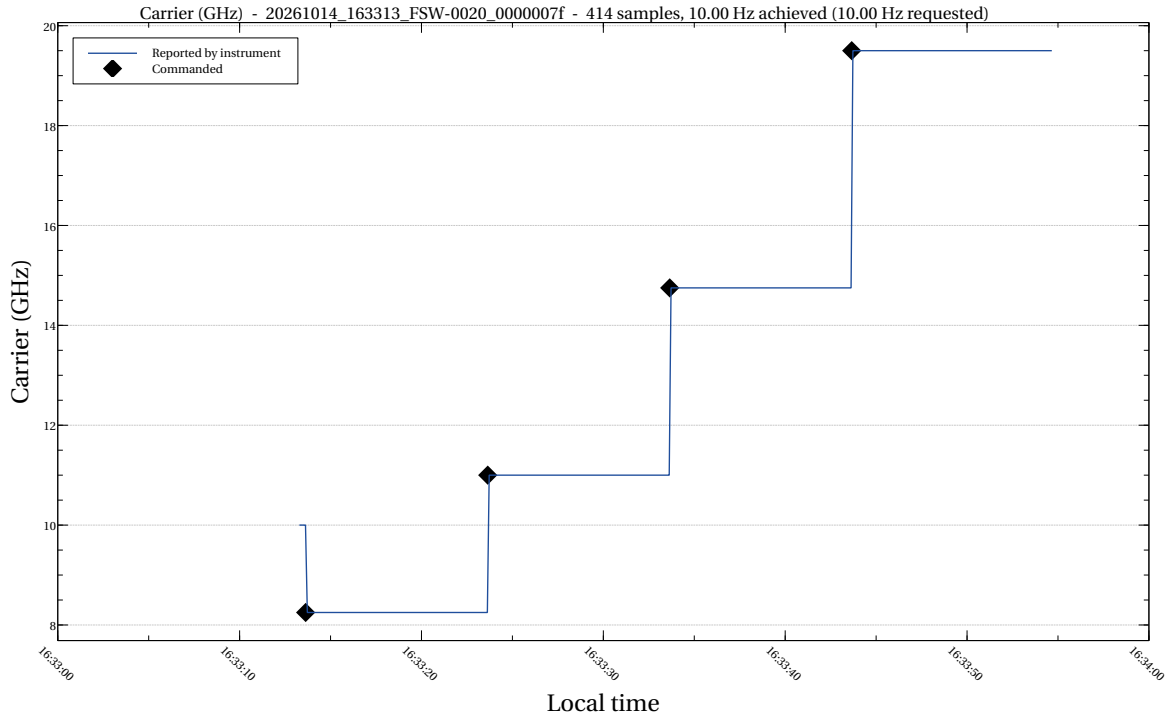
6 Plots

Every page of `quicksyn_session.pdf`, rendered by Veusz [5] from `quicksyn_session.vszh5`, included as vector graphics. Commanded values are drawn as black diamonds.

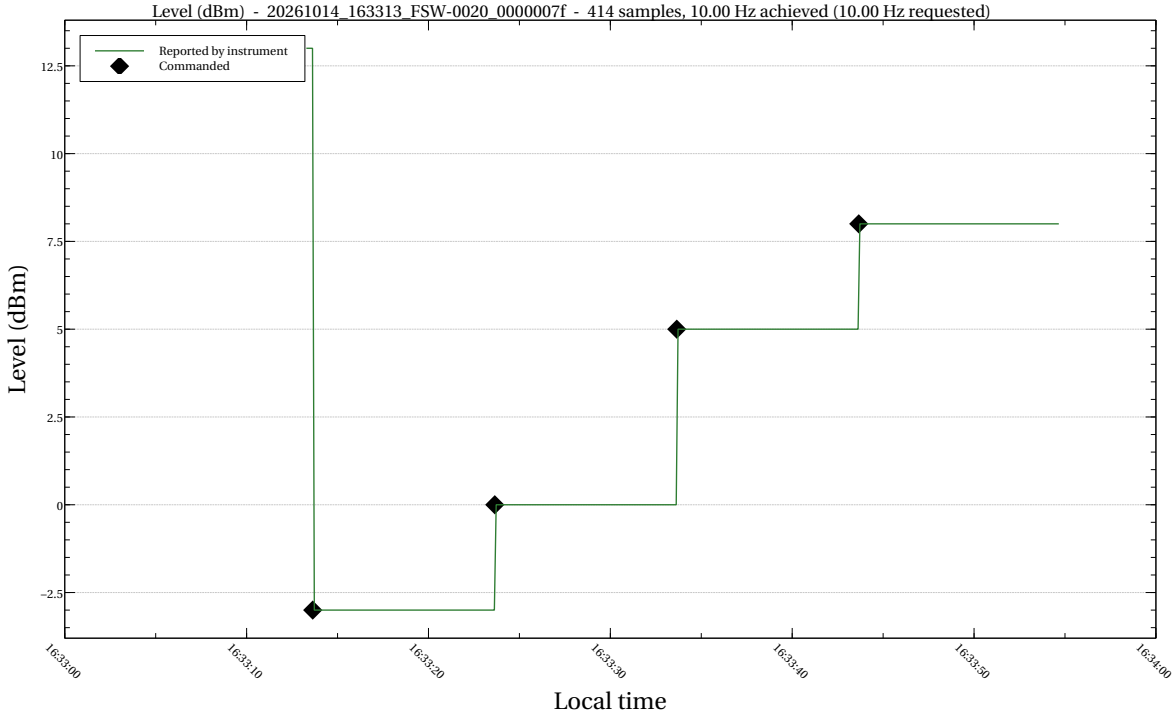
6.1 Overview (all quantities)



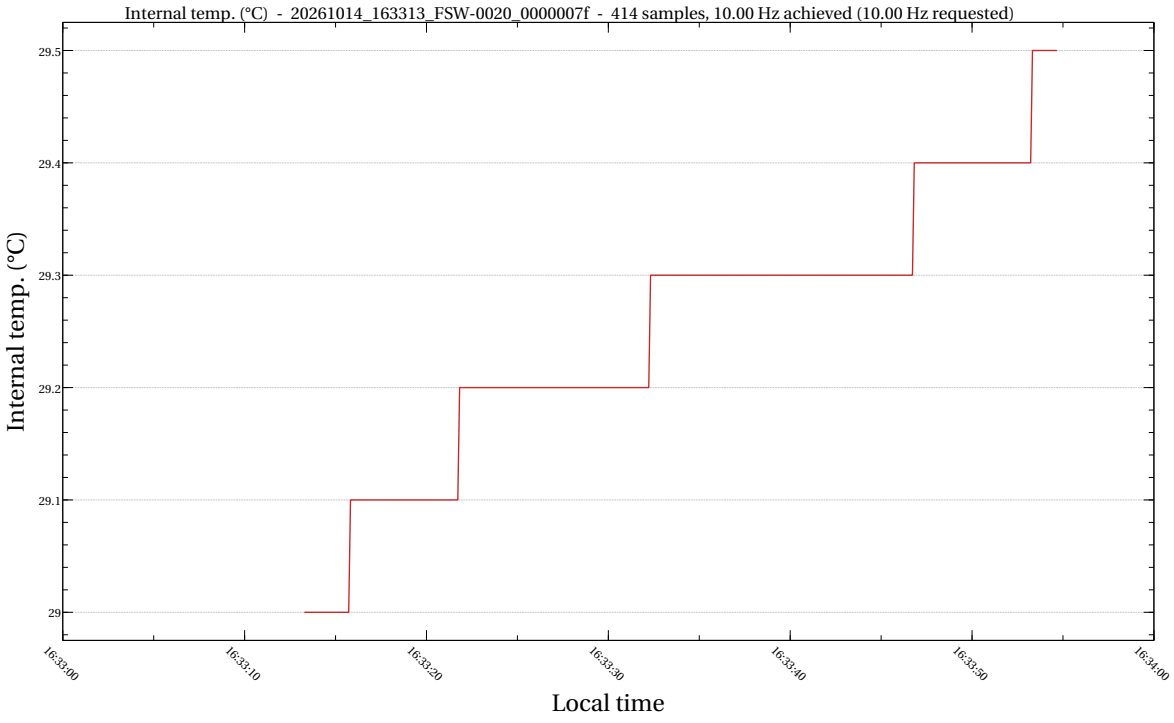
6.2 Carrier frequency



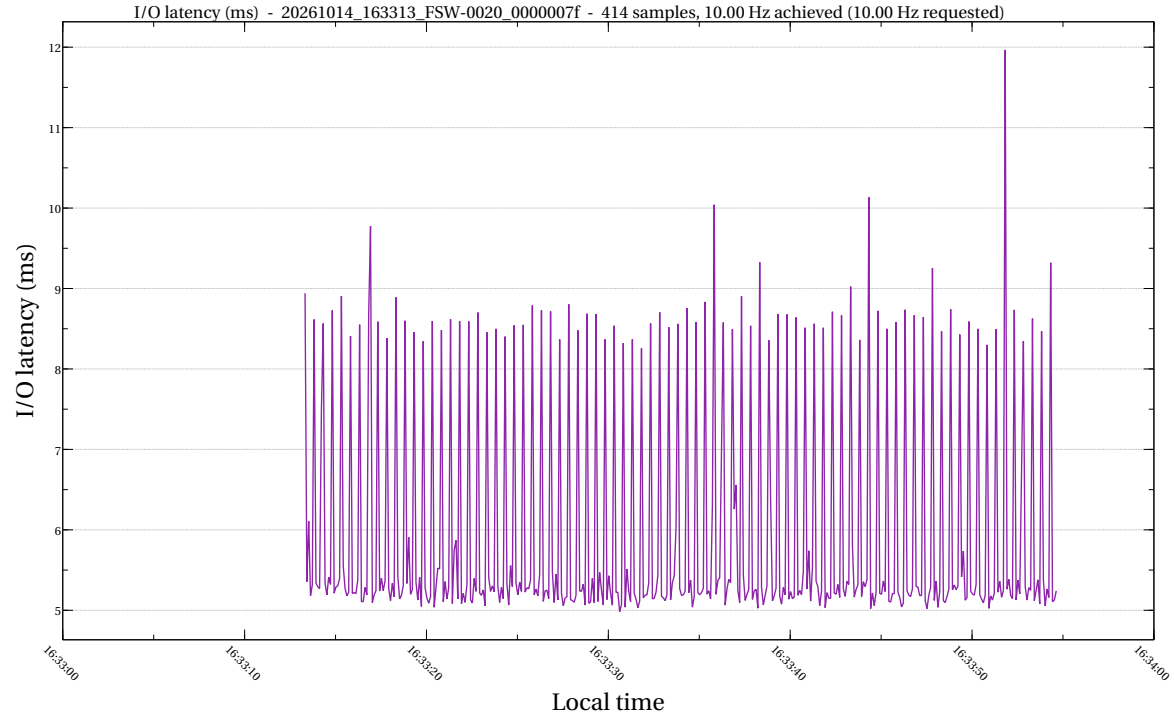
6.3 Output level



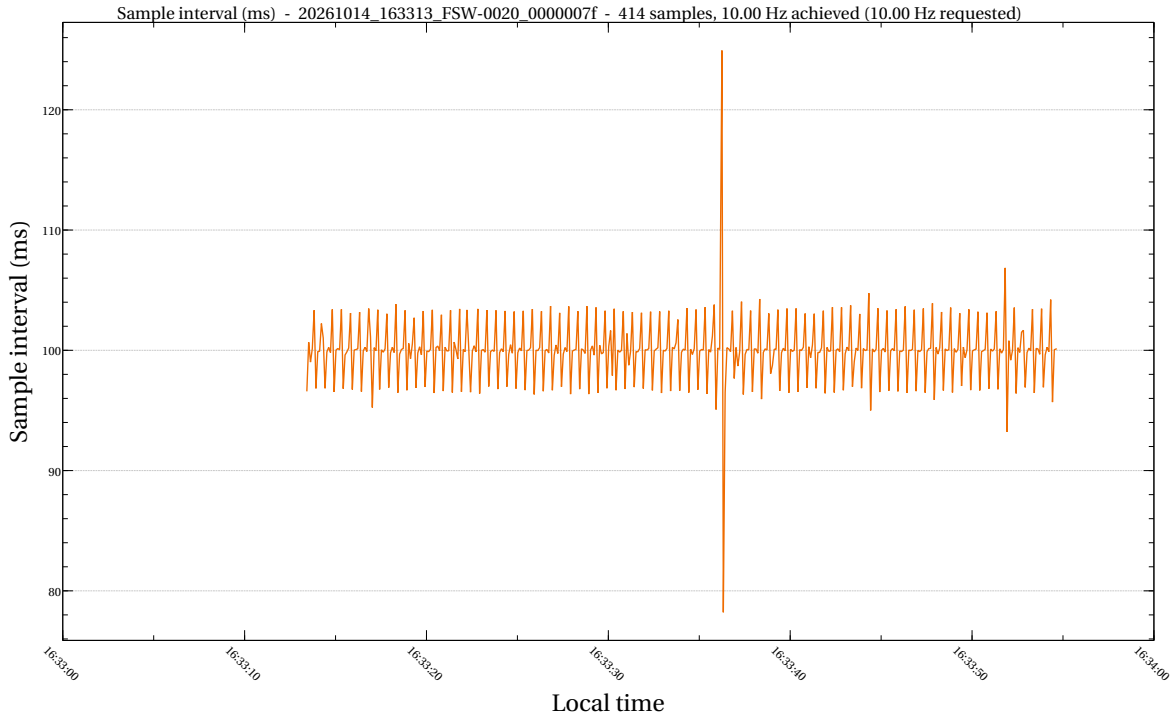
6.4 Internal temperature



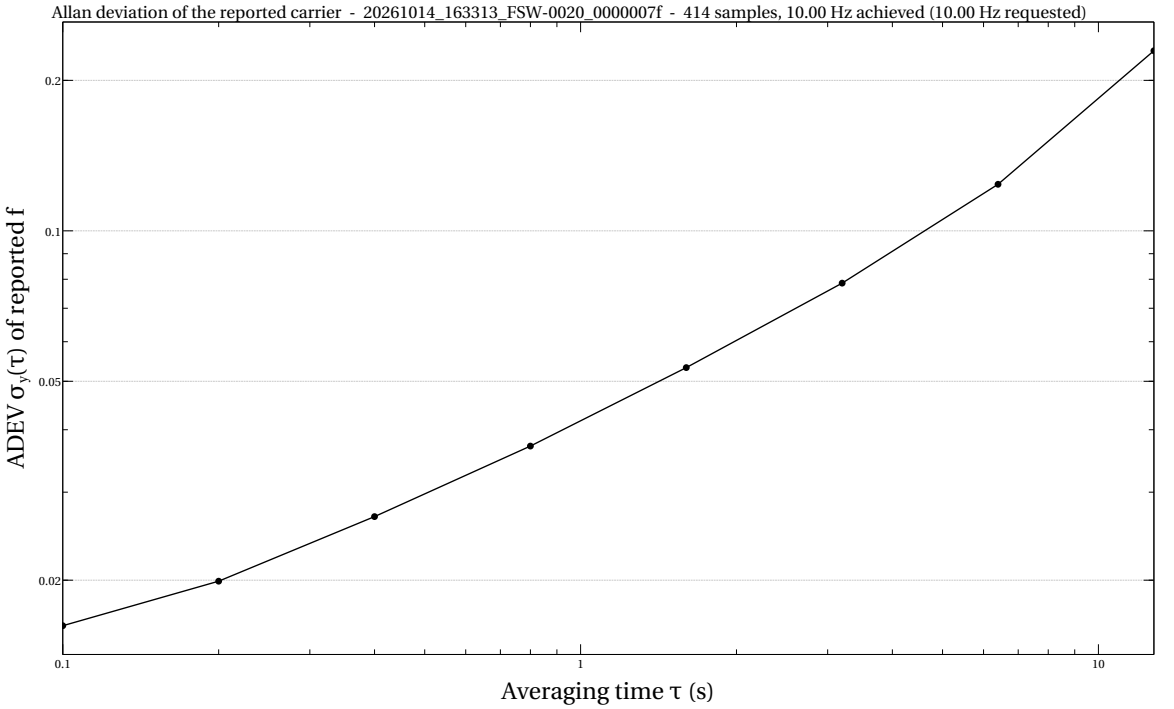
6.5 Serial I/O latency



6.6 Sample interval



6.7 Allan deviation of the reported carrier



7 Files in the session folder

- `quicksyn_session.pdf`
- `quicksyn_session.vszh5`
- `quicksyn_session_p00_page_overview.png`
- `reduction_summary.json`
- `reduction_summary.md`
- `session.log`
- `session_metadata.json`
- `state_events.csv`
- `telemetry.csv`
- `telemetry_chunk_0000.pkl.gz`
- `telemetry_reduced.npz`
- `transactions.csv`
- `quicksyn_session_report.tex` and `quicksyn_session_report.pdf` (this report)

8 Provenance

Statistics computed with the numpy backend in 8.7 ms by quicksyn_veusz_export_v0_0_10.py 0.0.10. The same numbers are in `reduction_summary.json` and `reduction_summary.md`.

References

[1] National Instruments, "QuickSyn Microwave Synthesizers: Communications Specifications," Doc. No. 5580510-01, Santa Clara, CA, USA, 2014.

[2] National Instruments, "QuickSyn Microwave Frequency Synthesizer: User Guide," Doc. No. 5580513-01, Rev. F.

[3] National Instruments, "QuickSyn Frequency Synthesizers," product brochure and datasheet, Oct. 2017.

[4] *IEEE Standard Definitions of Physical Quantities for Fundamental Frequency and Time Metrology—Random Instabilities*, IEEE Std 1139-2008, 2008. [Online]. Available: `https://ieeexplore.ieee.org/document/4797525`

[5] Veusz Developers, "Veusz—a scientific plotting package." [Online]. Available: `https://veusz.github.io/`